

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Project Report & Business Recommendations

Prepared for: Executive & Stakeholder Review

Scope: Data Preparation (Python) · Data Analysis (SQL) · Visualization (Power BI)

Report Date: September 2026

1. Executive Summary

Across 3,900 customer transactions, average spend per order is \$59.76 and the average product review is 3.75 out of 5. The single biggest opportunity in the data: 2,518 customers have made more than 5 purchases but are still not enrolled in the subscription program — a proven, high-value audience that is not being converted. Revenue also skews heavily by gender (68% male vs. 32% female) but is nearly flat across age groups, and five products (Hat, Sneakers, Coat, Sweater, Pants) rely on discounts for close to half their sales.

This report summarizes the end-to-end analysis: raw customer data was cleaned and modeled in Python, loaded into PostgreSQL, and queried in SQL to answer ten targeted business questions; results were then visualized in an interactive Power BI dashboard for stakeholder self-service. The sections below translate that technical work into plain-language findings, the underlying query results, and concrete next steps.

3,900	\$59.76	3.75 / 5	27%
Customers Analyzed	Avg. Purchase Amount	Avg. Review Rating	Subscribed to Loyalty Program

Table 1. Headline KPIs (source: Power BI dashboard)

2. Methodology

The analysis followed a three-layer pipeline: clean and model the raw data (Python), analyze it with structured queries (SQL/PostgreSQL), and present it for decision-makers (Power BI). Each layer is summarized below in plain language; full technical detail is in the Appendix.

2.1 Data Source

The source dataset is a customer shopping behavior file (customer_shopping_behavior.csv) containing one row per transaction, with fields covering customer demographics (age, gender), the item purchased and its category, purchase amount, review rating, discount/promo usage, subscription status, shipping type, payment method, and purchase frequency.

2.2 Data Preparation (Python)

Using pandas, the raw data was inspected and cleaned before analysis:

- Missing review ratings were filled with the median rating for that product's category, rather than a single global average, to avoid skewing ratings across very different product types.
- Column names were standardized to lowercase with underscores, and the purchase amount column was renamed for consistency (purchase_amount).
- Customers were split into four equal-sized age_group bands (Young Adult, Adult, Middle-aged, Senior) using quantile binning, so each group represents a comparable share of the customer base.
- A purchase_frequency_days field was derived from the text frequency labels (e.g., "Weekly" → 7, "Monthly" → 30, "Annually" → 365) to make purchase cadence usable in calculations.

- The `promo_code_used` column was dropped after confirming it was identical to `discount_applied` in every row, removing redundant data.
- The cleaned dataset was loaded into a PostgreSQL database (customer table) using SQLAlchemy, making it queryable with SQL for the analysis phase.

2.3 Data Analysis (SQL)

Ten business questions were translated into SQL queries against the customer table — covering revenue by gender, discount behavior, top-rated products, shipping method comparisons, subscriber value, customer segmentation (New / Returning / Loyal), top products per category, repeat-buyer loyalty likelihood, and revenue by age group. The full query set is provided in Appendix A.

2.4 Visualization (Power BI)

Query outputs and KPIs were connected to an interactive Power BI dashboard with slicers for subscription status, gender, product category, and shipping type, so stakeholders can filter the view themselves without needing to write queries. The dashboard is reproduced in Section 3.

3. Key Findings

The figures below combine the Power BI dashboard with the actual SQL query outputs run against the PostgreSQL customer table (full result screenshots are in Appendix A). Where the dashboard shows a visual summary, the SQL layer confirms the exact underlying numbers.

3.1 Dashboard Overview



Figure 1. Customer Behavior Dashboard (Power BI)

3.2 Revenue Skews Heavily Male

Gender	Total Revenue	Share of Revenue
Male	\$157,890	68%
Female	\$75,191	32%

Table 2. Revenue by gender (Q1 query result)

Male customers generate roughly double the revenue of female customers. The dataset doesn't tell us whether this is driven by more male customers, higher spend per male customer, or both — that split is worth pulling next — but as-is, gender is currently the single sharpest revenue divide in the data, more pronounced than any age-group difference.

3.3 Revenue and Sales by Age Group

Age Group	Total Revenue
Young Adult	\$62,143
Middle-aged	\$59,197

Adult	\$55,978
Senior	\$55,763

Table 3. Revenue by age group (Q10 query result)

Revenue is tightly clustered between \$55.8K and \$62.1K across all four age bands — only an 11% spread from lowest to highest. Combined with the gender split above, this confirms the business has broad, age-agnostic appeal, but a much sharper split by gender.

3.4 Subscription Program Is Under-Delivering

Status	Customers	Avg. Spend	Total Revenue
Subscribed (Yes)	1,053	\$59.49	\$62,645
Not Subscribed (No)	2,847	\$59.87	\$170,436

Table 4. Subscriber vs. non-subscriber spend (Q5 query result)

Only 27% of customers (1,053 of 3,900) are subscribed — and subscribers actually spend slightly less per order on average (\$59.49) than non-subscribers (\$59.87). The subscription program is not currently correlated with higher basket size, which means its value has to come from retention and repeat-purchase frequency rather than order value — and that link isn't yet proven in this data.

3.5 Repeat Buyers Are Not Being Converted to Subscribers

Subscribed?	Repeat Buyers (>5 previous purchases)
No	2,518
Yes	958

Table 5. Repeat buyers by subscription status (Q9 query result)

This is the clearest opportunity in the dataset: of 3,476 customers with more than 5 previous purchases, 72% (2,518) are still not subscribed. These are proven, repeat shoppers with no formal loyalty relationship — the highest-value, lowest-effort segment to target for subscription conversion.

3.6 Customer Segments Skew Loyal

Segment	Number of Customers	Share
Loyal (>10 purchases)	3,116	80%
Returning (2–10 purchases)	701	18%
New (1 purchase)	83	2%

Table 6. Customer segmentation (Q7 query result)

80% of the customer base already qualifies as "Loyal" by purchase count, and only 2% are brand-new, single-purchase customers. Retention efforts have clearly worked — the more pressing question is top-of-funnel: with so few New customers in the mix, the business may be under-acquiring rather than under-retaining.

3.7 Top-Rated and Discount-Reliant Products

The five highest-rated products (Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78) are all reviewed just under 4.0 — solid, but with headroom. Notably, Hat also tops the discount-dependency list at a 50% discount-purchase rate, alongside Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) — meaning close to half of these products' sales rely on a discount being applied. Full result tables are in Appendix A (Q3, Q6).

3.8 Shipping and Discount Behavior

Express shipping customers spend marginally more on average (\$60.48) than Standard shipping customers (\$58.46) — a small but consistent gap. Separately, 839 customers used a discount and still spent above the \$59.76 average purchase amount, confirming a meaningful "deal-seeking but high-value" segment rather than discounts only attracting low-spend, price-sensitive shoppers. Full result tables are in Appendix A (Q2, Q4).

4. Insights & Recommendations

4.1 Convert Repeat Buyers Into Subscribers — Highest-Priority Action

2,518 customers have bought more than 5 times but are still not subscribed (Section 3.5). This is a warm, low-effort audience.

- Target these 2,518 customers directly with a subscription invitation — they've already shown repeat intent, so the pitch is retention benefits, not first-time trust-building.
- Since subscribers don't currently spend more per order (Table 4), reposition the program's value around perks, early access, or free shipping rather than implying it will raise basket size.

4.2 Investigate the Gender Revenue Gap

Male customers drive 68% of revenue versus 32% for female customers (Section 3.2) — the widest gap in the dataset.

- Pull customer counts by gender to determine whether this is a spend-per-customer gap or a customer-count gap; the fix is different for each.
- If it's a count gap, prioritize acquisition marketing toward female customers; if it's a spend gap, test bundling or upsell offers on the categories female customers already buy.

4.3 Fix the Discount-Dependent Products

Hat, Sneakers, Coat, Sweater, and Pants each sell on discount roughly half the time (Section 3.7).

- Test a modest price reduction or a bundle offer on these five products in place of the recurring discount, to see if full-price demand exists without training customers to wait for a deal.
- Cross-merchandise these items with top-rated products (Gloves, Sandals, Boots) to borrow trust and reduce reliance on price as the primary lever.

4.4 Address the Acquisition Gap

Only 83 of 3,900 customers (2%) are "New" by segment definition (Section 3.6), while 80% are already "Loyal."

- Audit top-of-funnel marketing spend — a customer base this skewed toward existing loyal buyers suggests new-customer acquisition may be underfunded relative to retention.
- Use the top-rated and top-selling products per category (Appendix A, Q3 and Q8) as the entry point for new-customer campaigns, since they carry the strongest social proof.

4.5 Keep Marketing Broad by Age

Revenue is nearly flat across all four age groups (Section 3.3, 11% spread), so age-based segmentation is unlikely to be a high-leverage lever compared to the gender and loyalty-status splits above.

5. Appendix

A. SQL Query Results

Each query below was run against the live PostgreSQL customer table. Screenshots show the actual output; the full SQL text for each is available in Customer_behavior_business_Analysis.sql in the project repository.

Q1. Total revenue by gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Finding: Male customers generate \$157,890 vs. \$75,191 for female customers — a 68/32 split.

Q2. Discount users who still spend above average

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	10	70

Finding: 839 customers used a discount and still spent above the \$59.76 average purchase amount (first 10 rows shown).

Q3. Top 5 products by average review rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Finding: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78) are the top-rated products.

Q4. Standard vs. Express average purchase amount

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Finding: Express shipping customers spend slightly more on average (\$60.48) than Standard shipping customers (\$58.46).

Q5. Subscriber vs. non-subscriber spend

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Finding: Subscribers (1,053 customers) spend marginally less per order (\$59.49) than non-subscribers (\$59.87), and contribute far less total revenue.

Q6. Products with the highest discount-purchase rate

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Finding: Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) rely on discounts for close to half their sales.

Q7. Customer segmentation: New / Returning / Loyal

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Finding: 80% of customers (3,116) are already "Loyal," only 2% (83) are "New."

Q8. Top 3 products per category

	bigint	category text	product text	count bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Finding: Best-sellers by category: Jewelry, Sunglasses, Belt (Accessories); Blouse, Pants, Shirt (Clothing); Sandals, Shoes, Sneakers (Footwear); Jacket, Coat (Outerwear).

Q9. Repeat buyers and subscription likelihood

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Finding: Of 3,476 customers with more than 5 previous purchases, 72% (2,518) are still not subscribed.

Q10. Revenue contribution by age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Finding: Revenue is tightly clustered: Young Adult \$62,143, Middle-aged \$59,197, Adult \$55,978, Senior \$55,763.

B. Python Data Preparation (Key Steps)

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')

# Fill missing review ratings with the category median
df['Review Rating'] = df.groupby('Category')['Review Rating']\
    .transform(lambda x: x.fillna(x.median()))

# Standardize column names
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

# Build age_group via quantile binning
labels = ['Young Adult', 'Adult', 'Middle-aged', 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

# Convert purchase frequency text to days
frequency_mapping = {
    'Fortnightly': 14, 'Weekly': 7, 'Monthly': 30, 'Quarterly': 90,
    'Bi-Weekly': 14, 'Annually': 365, 'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

# Drop redundant column (identical to discount_applied)
df = df.drop('promo_code_used', axis=1)

# Load into PostgreSQL for SQL analysis
from sqlalchemy import create_engine, URL
url = URL.create('postgresql+psycopg2', username=username, password=password,
                host=host, port=port, database=database)
engine = create_engine(url)
df.to_sql('customer', engine, if_exists='replace', index=False)
```

Code has been lightly reformatted for readability; credentials are placeholders (see Note below).

C. Assumptions & Limitations

- Category revenue in Figure 1 (Clothing/Accessories/Footwear/Outerwear) is a dashboard visual only — no corresponding SQL query was included in the provided script, so those figures are approximate, read from the chart axis rather than an exact query result.

- Q1 (revenue by gender) reports total revenue only, not customer counts per gender — the recommendation in Section 4.2 to investigate a count-gap vs. spend-gap requires an additional query not included in the original script.
- Age groups are quantile-based (equal customer counts per band), not fixed age ranges — comparisons across groups reflect relative position within this customer base, not standardized age brackets.
- The notebook's database connection cell contains a hard-coded username and password; these should be moved to environment variables or a secrets manager before the code is shared or committed to a repository.
- "promo_code_used" was dropped after confirming it matched "discount_applied" in every row of the sampled data — this should be re-verified if the source data is refreshed.